RE6

Kinetics: Heterogeneous Catalyzed Reaction

Methanation of CO_2 over a Ru/Al_2O_3 catalyst

TKP 4110 - Reaksjonssteknikk
March 19, 2026

Group B 21
Tuva Høven, Sigurd Jensen and Sigrid Haugneland Christensen

Contents

1 Introduction **1**

2 Theory **1**
2.1 Reaction kinetics 1
2.2 Gas Chromatography 2

3 Experimental **3**
3.1 Preparation 3
3.2 Calibration 3
3.3 Catalyst Pretreatment 4
3.4 Kinetic Experiment 4
3.5 System shutdown 4

A Calculations **6**
A.1 Rate of Consumption 6
A.2 Reaction Order 6
A.3 Activation Energy 6
A.4 CO_2 Conversion 6
A.5 Molar Flow Rates and MFC Input Signal 7
A.6 BEREGNINGER TIL PRESENTASJON 7

B HSE **8**

1 Introduction

The CO_2 methanation reaction will be studied over a Ru-based catalyst under varying conditions. It will take place in a fixed-bed quartz reactor at 260–300°C with atmospheric pressure and H_2-rich conditions. The reactant stream contains CO_2, H_2, He, and N_2 (internal standard), with the effluent analyzed via micro-GC. Calibration of mass-flow controllers and a three-level micro-GC calibration will be conducted. The reaction order of CO_2 (using a power-law model) and activation energy (via the Arrhenius equation) will be determined by adjusting CO_2 partial pressure and temperature.

2 Theory

2.1 Reaction kinetics

I order to obtain a sustainable CH_4 production it has been proposed to use green H_2 from water hydrolysis with electricity from renewable energy sources. The conversion of CO_2 and H_2 to CH_4 is described by the exothermic reaction 2.1. The reaction occurs on solid catalysts containing either Co, Ni, Fe or Ru. If the conditions are met the side reaction *reverse water gas shift* (RWGS) may happen. The side reaction 2.2 is endothermic. [1]

$$CO_2 + 4\,H_2 \rightleftharpoons CH_4 + 2\,H_2O \tag{2.1}$$

$$CO_2 + 4\,H_2 \rightleftharpoons CO + H_2O \tag{2.2}$$

Equation 2.3 is the power law that can describes the kinetics of the reaction 2.1 at low conversion levels. $-r_{CO_2}$ is the rate consumption of CO_2, k is the rate constant,p is the partial pressure of the reactants and a and b are the respective reaction orders of CO_2 and H_2. The reaction rate for H_2 can be considered independent from the partial pressure for H_2-rich atmospheres *(b = 0)*. Therefore equation 2.3 can be reduced to equation 2.4. [1]

$$-r_{CO_2} = kp^a_{CO_2} p{H_2}^b \tag{2.3}$$

$$-r_{CO_2} = kp^a_{CO_2} \tag{2.4}$$

At differential reactor conditions, the reaction rate is related to the conversion level by equation 2.5. The conversion level of CO_2 is required to be sufficiently low for the differential reactor assumption to be valid. This is because the CO_2 partial pressure can be considered constant throughout the reactor. However, if the conversion were high, the changing CO_2 partial pressure would need to be accounted for, requiring a more detailed integral reactor model. [1]

$$-r_{CO_2} = \frac{F^0_{CO_2} X_{CO_2}}{\Delta W} \tag{2.5}$$

$F^0_{CO_2}$ is the molar flow rate of CO_2 into the reactor, X_{CO_2} is the CO_2 conversion and ΔW is the applied catalyst mass.

The reaction order *(a)* of CO_2 can be determined by plotting the equation 2.6. This will linearize the function and the reaction order a can be calculated from the slope by linear regression. [1]

$$ln(-r_{CO_2}) = ln(k) + a \cdot ln(p_{CO_2}) \tag{2.6}$$

When the CO_2 reaction order a is determined the rate constant k at different reaction temperatures can be found experimentally based on equation 2.4. The rate constant k is also related to the reaction temperature by the Arrhenius equation 2.7. The activation energy E_a can be determined by linearization of equation 2.7 as shown in equation 2.8. [1]

$$k = Ae^{-\frac{E_a}{RT}} \tag{2.7}$$

Where A is the pre-exponential factor, R is the universal gas constant and T is the reaction temperature.

$$ln(k) = ln(A) - \frac{E_a}{RT} \tag{2.8}$$

An Arrhenius diagram describes a lot of the properties of the system. The diagram can be achieved by plotting $\frac{1}{T}$ against the natural logarithm of k, calculated from equation 2.8. The slope of the linear regression is $\frac{E_a}{R}$ and A would be the y-axis interception value.

2.2 Gas Chromatography

Gas chromatography (GC) is a method that separates compounds in a gas mixture for analysis and can be used for both qualitative and quantitative purposes. An inert carrier gas (mobile phase) is used to transport an injected gas phase through the system [2]. The stationary phase consists of either a solid or liquid coating on the inside of the GC columns. As the gas moves through the column, the different compounds interact with the stationary phase differently, which results in their separation. This separation is dependent on the compounds' different affinities (attraction between the compounds and the stationary phase) [3]. Compounds with high affinity move slower through the column than those with lower affinity [4]. Consequently, the compounds exit the column at different times, known as retention times. When leaving the column, the compounds enter a detector.

Micro-GC is a high-speed GC system used for rapid analyses of gas mixtures. A thermal conductivity detector (TCD) is often used in a micro-GC, which is a detector that measures differences in thermal conductivity between the gases exiting the GC column and a reference carrier gas (inert).

The GC needs to be calibrated with relevant gases to function as both a qualitative and quantitative analysis tool. To do this, gas mixtures with known compositions are analyzed in order to identify the retention time and response factor (RF_i) for each component. Eq. (2.9) is used to determine the RF_i.

$$RF_i = \frac{A_i}{X_i} \tag{2.9}$$

Here, A_i and X_i are the peak integration area and the calibrated molar fraction.

The internal standard method is a commonly used calibration method, where the same amount of an inert gas (internal standard) is added to each sample. The amount of the other components are based relative to the amount of the inert gas. In this experiment, N_2 is used as the internal standard. Eq. (2.10) shows how to calculate the relative response factor, which is a measure of how the response of a detector for a given compound compares to the response of the internal standard.

$$\begin{aligned} RRF_i &= \frac{RF_i}{RF_{IS}} \\ &= \frac{A_i}{x_i} \cdot \frac{x_{IS}}{A_{IS}} \end{aligned} \tag{2.10}$$

Here, RF_{IS} is the response factor for the internal standard, where A_{IS} and x_{IS} are the peak area and molar fraction for the internal standard. RF_i, A_i, and x_i are the response factor, peak area, and molar fraction for component i.

By multiplying the right side of Eq. (2.10) by $\frac{F_{tot}}{F_{tot}}$, an equation for the RRF_i is obtained as a function of the response factors and the peak areas, as shown in Eq. (2.11).

$$\begin{aligned} RRF_i &= \frac{A_i}{A_{IS}} \cdot \frac{x_{IS}}{X_i} \cdot \frac{F_{tot}}{F_{tot}} \\ &= \frac{A_i}{A_{IS}} \cdot \frac{F_{IS}}{F_i} \end{aligned} \tag{2.11}$$

This can be rearranged to get an equation for $\frac{A_i F_{IS}}{A_{IS}}$, shown in Eq. (2.12).

$$\frac{A_i F_{IS}}{A_{IS}} = F_i \cdot RRF_i \tag{2.12}$$

With Eq. (2.12), it is possible to calculate the RRF_i by using the GC to find the response factors and the peak areas, and then plot $\frac{A_i F_{IS}}{A_{IS}}$ as a function of different F_i values. This plot should be linear, with the RRF_i as the slope.

3 Experimental

3.1 Preparation

- Load the reactor with 100 mg Ru/Al_2O_3.
- Dilute the catalyt bed with 900 mg inert SiC and place a layer above and below the catalyst bed to stabilize it.
- The reactor is then closed and placed inside the heating furnace and connect the gas-feed and outlet lines.
- Place the thermocouple inside the thermocouple holder in the reactor center.
- Leak test the N_2-gas using a leak detection spray, then H_2-gas with a flammable gas detector.
- Purge the system with N_2-gas for 15 minutes through the reactor, and then 15 minutes through the reactor bypass.
- Introduce the N_2-gas to the system and turn on the condencer cooling water.
-

3.2 Calibration

- Set the to direct the flow through the reactor bypass
- Introduce the gasses one by one while analyzing the flow with the micro-GC to identify the peak position of each component in the chromatogram. The time it takes to reach the peak position is the retention time.
- Do a three-level calibration of the micro-GC with the conditions in table 3.1.
- Determine the relative response factor of CO_2 with respect to the internal standard N_2 by following the procedure in subsection 3.1.
- Do three measurements at each calibration level.

Table 3.1: Experimental conditions for calibration and the micro-GC

Calibration Level	*Volume Fraction [vol%] (a)*				Total flow [mL/min] (a)
	CO_2	H_2	N_2	He	
1	5	60	15	20	200
2	7.5	60	15	17.5	200
3	10	60	15	15	200

3.3 Catalyst Pretreatment

- Set the three-valves to direct the flow through the reactor
- Reduce the Ru-catalyst by using 50 vol% H_2 in N_2 with a total flow of 200 mL/min while heating the catalyst bed to 350 °C. The heating is done by an automated temperature program

3.4 Kinetic Experiment

-
- Determine the activation energy by adjusting the system to the three first steps in table 3.2
- Determine the CO2 reaction order by adjusting the system to the two last steps in table 3.2
- Adjust the N_2, H_2 and He flows before introducing CO_2 and manually adjust the temperature.
- Only the measurements obtained under steady-state conditions are being used (at least 3 with each reaction conditions). It may take up to 30 minutes to reach steady-state.

Table 3.2: Reaction conditions for the kinetic experiment

Step	Volume Fraction [vol%] (a)				Total flow [mL/min] (a)	Temperature
	CO2	H2	N2	He		
1	5	60	15	20	200	300
2	5	60	15	20	200	280
3	5	60	15	20	200	260
4	7.5	60	15	17.5	200	260
5	10	60	15	15	200	260

3.5 System shutdown

- Close the H2, He and CO2 gas flows.
- Turn off the heating furnace
- Purge the system with N2 (100 mL/min) while the system cools down.
- Leave the cooling water running until the system reaches room temperature.

References

[1] H. Scott Fogler. *Elements of Chemical Reaction Engineering.* Pearson, London, England, 6th edition, 2020.

[2] et.al Tarsila Rodrigues Arruda. *Green Products in Food Safety.* Elsevier Inc, 2023.

[3] Bjørn Pedersen. affinitet. `https://snl.no/affinitet`, 2024. Hentet 09.10.2024.

[4] Grethe Wibetoe. gasskromatografi. `https://snl.no/gasskromatografi`, 2022. Hentet 09.10.2024.

[5] ROTH. Frivillig sikkerhetsinformasjon basert på sikkerhetsbladformat iht. forordning (EF) nr. 1907/2006 (REACH). `https://shorturl.at/bBuTC`, 2017. Hentet 09.10.2024.

A Calculations

A.1 Rate of Consumption

$-r_{CO_2}$ is the rate of CO_2-consumption, and is given by

$$\begin{aligned} -r_{CO_2} &= kp_{CO_2}^a p_{H_2}^b \\ &= kp_{CO_2}^a, \end{aligned} \tag{A.1}$$

where k is the rate constant, p_{CO_2} is the partial pressure for CO_2, p_{H_2} is the partial pressure for H_2, a is the reaction order for CO_2, and b is the reaction order for H_2. In a H_2-rich atmosphere, the reaction constant is considered independent of the partial pressure of H_2, hence the simplification of $-r_{CO_2}$ in A.1.

When assuming differential reactor conditions, the consumption of CO_2 at low conversion levels, can be expressed as

$$-r_{CO2_2} = \frac{F_0^{CO_2} X_{CO_2}}{\Delta W} \tag{A.2}$$

where X_{CO_2} is the conversion level for CO_2, $F_0^{CO_2}$ is the molar flow rate of Co_2 in the inlet of the reactor, and ΔW is the catalyst sample mass.

A.2 Reaction Order

To find the reaction rate and reaction order, equation A.1 can be rewritten as

$$ln(-r_{CO2_2}) = lnk + \alpha \cdot ln(p_{CO_2}) \tag{A.3}$$

where k ais the rate constant, and α is the reaction order. α, can be found by plotting the natural logarithm of the reaction rate, $ln(r_{CO_2})$, as a function of p_{CO_2} at different partial pressures for CO_2. The reaction order is given by the value pf the slope, and the rate constant is the functions point of intercetpt with the y-axis.

A.3 Activation Energy

By rewriting the Arrhenius equations, the activation energy, E_A, can be found

$$k = Ae^{\frac{-E_A}{RT}} \tag{A.4}$$

$$lnk = lnA - \frac{-E_A}{RT} \tag{A.5}$$

where A is the constant pre-exponential factor, R is the gas constant, and T is the reactor's temperature. By plotting lnk as a function of $\frac{1}{T}$, $\frac{E_A}{R}$ can be found as the slope value, and lnA as the intercept value.

A.4 CO_2 Conversion

To find the CO_2 conversion, gas vhromatography (GC) will be used. The response factor for component i, RF_i, can be found by using the following equation

$$RF_i = \frac{A_i}{x_i} \tag{A.6}$$

where A_i is the peak integration, and x_i is the molar fraction.

A.5 Molar Flow Rates and MFC Input Signal

The flow rates and MFC input signals in table **??** are found by using equations A.7-A.10.

$$S_{H_2} = 2.73 \cdot F_{H_2} - 6.56 \tag{A.7}$$

$$S_{CO_2} = 0.2334 \cdot F_{CO_2} - 1.7006 \tag{A.8}$$

$$S_{N_2} = 2.19 \cdot F_{N_2} + 1.57 \tag{A.9}$$

$$S_{He} = 2.11 \cdot F_{He} - 0.40 \tag{A.10}$$

Calibration data for each component was given in the experimental description. To estimate the equations, the data was used, and the MFC set points, S_i, were plotted as a function of the flow rates, F_i. The plot was found using linear regression.

Table A.1: Mekanisme for en elektrofil aromatisk substitusjon

		MFC Setpoint [%] (a)				Volume Flow [mL/min] (b)				Volume Fraction [vol%] (b)				Total Flow [mL/min] (b)	Temperature [°C]	Reactor Bypass
		CO_2	H_2	N_2	He	CO_2	H_2	N_2	He	CO_2	H_2	N_2	He			
GC Calibration	Level 1	49.6	46.7	13.2	18.6	10	120	30	40	5	60	15	20	200	RT	Yes
	Level 2	70.9	46.7	13.2	16.2	15	120	30	35	7.5	60	15	17.5	200	RT	Yes
	Level 3	93.7	46.7	13.2	13.9	20	120	30	30	10	60	15	15	200	RT	Yes
Reduction		0	39.3	44.6	0	0	100	100	0	0	50	50	0	200	RT → 350 °C → 300 °C	No
Kinetic Experiment	Step 1	49.6	46.7	13.2	18.6	10	120	30	40	5	60	15	20	200	300 °C	No
	Step 2	49.6	46.7	13.2	18.6	10	120	30	40	5	60	15	20	200	280 °C	No
	Step 3	49.6	46.7	13.2	18.6	10	120	30	40	5	60	15	20	200	260 °C	No
	Step 4	70.9	46.7	13.2	16.2	15	120	30	35	7.5	60	15	17.5	200	260 °C	No
	Step 5	93.7	46.7	13.2	13.9	20	120	30	30	10	60	15	15	200	260 °C	No
Cool-down		0	0	90.5	0	0	0	200	0	0	0	100	0	100	260 °C → RT	No

A.6 BEREGNINGER TIL PRESENTASJON

Step	A_{N_2} [bar]	A_{CO_2} [K]	p_{CO_2} [mol/min]	T [%]	F_{CO_2} [mol/min g]	X_{CO_2}	$-r_{CO_2}$	k
1.1	2784.641	11779.996	1.783×10^{-4}	7.62	7.35×10^{-5}	-	-	
1.2	2936.548	11440.618	0.05	623	1.733×10^{-4}	10.21	9.85×10^{-5}	-
1.3	3315.983	11344.289	1.715×10^{-4}	11.14	1.08×10^{-4}	-	-	
2.1	2888.322	15464.501	2.305×10^{-4}	20.51	2.97×10^{-4}	-	-	
2.2	3213.371	15669.937	0.075	623	2.331×10^{-4}	19.62	2.85×10^{-4}	-
2.3	3272.713	15311.7540	2.722×10^{-4}	21.66	3.14×10^{-4}	-	-	
3.1	3563.179	20963.936	3.068×10^{-4}	20.52	3.96×10^{-4}	0.0737	-	
3.2	3594.748	20548.656	0.10	623	2.992×10^{-4}	22.49	4.34×10^{-4}	0.0808
3.3	3702.941	20682.976	3.010×10^{-4}	22.02	4.25×10^{-4}	0.0791	-	
4.1	5321.007	19037.989	2.164×10^{-4}	31.21	6.31×10^{-4}	0.1175	-	
4.2	5351.128	18909.289	0.10	633	2.579×10^{-4}	32.13	6.10×10^{-4}	0.1136
4.3	5298.395	18767.110	2.556×10^{-4}	32.74	6.22×10^{-4}	0.1158	-	
5.1	6798.13	16657.563	2.106×10^{-4}	43.68	8.17×10^{-4}	0.1521	-	
5.2	69	15925.869	0.10	643	1.981×10^{-4}	47.04	8.80×10^{-4}	0.1639
5.3	66401.177	15716.061	1.946×10^{-4}	47.97	8.97×10^{-4}	0.1670	-	

B HSE

The gases used in this experiment are CO_2, N_2, H_2 and He. All of these come in containers that contains gas under pressure and can explode if it is heated. The H_2 gas is also highly flamabel, and therefore is a higher risk than the other gases. The use of personal protective equipment such as the use of protective gloves when handling the chemicals, along with a lab coat, goggles and closed toe shoes, is recommended.

Glass wool is utilized in this experiment and is composed of small, thin fibers that can irritate the respiratory tract when inhaled, potentially leading to coughing and respiratory discomfort. It is therefore advisable to wear a dust mask when handling it. [5]